Department Management – Technical Documentation

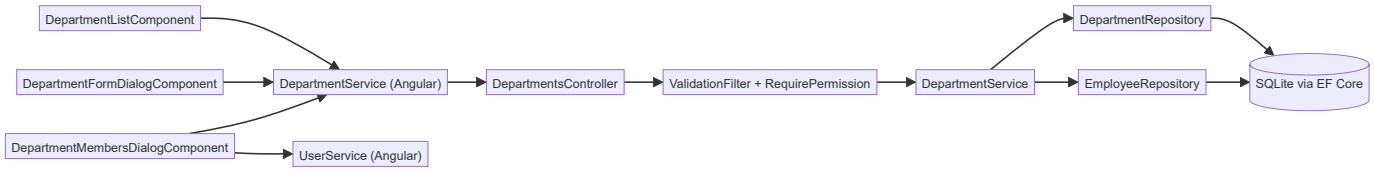
> **Project:** TaskFlow – Intelligent Task Allocation & Tracking System **Parent Module:** Management · **Child Module:** 03.2 Department Management **Version:** 1.0 · **Last Updated:** 31-Oct-2026 **Backend:** ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core) **Author:** Jojo George (jojogeorge3344@gmail.com) © 2026 Jojo George. All rights reserved.

1. Technical Overview

Department Management owns the `Departments`, `DepartmentManagers` and `UserDepartments` tables. It is served by `DepartmentsController` on `api/v1/departments` and implemented by `DepartmentService`, which uses `DepartmentRepository` for the department rows and `EmployeeRepository` to check the people picked as members.

- Every endpoint is protected by `[RequirePermission(PermissionKeys.X)]`.
- Departments are soft-deleted (`IsDeleted`) and only when they have no members. A global query filter hides deleted departments and the membership rows that point at them.
- The members endpoint replaces all managers and employees of one department in one call, diffing only that department's link rows.
- The frontend is the lazy route `/departments` with `DepartmentListComponent`, `DepartmentFormDialogComponent` and `DepartmentMembersDialogComponent`, calling the Angular `DepartmentService`.

2. Architecture Flow



3. Backend File Structure

Layer	Path	Files
Controller	`TaskFlow.Api/Controllers/Management/`	`DepartmentsController.cs`
Service	`TaskFlow.Api/Services/Management/DepartmentManagement/`	`IDepartmentService.cs`, `DepartmentService.cs`
Repository	`TaskFlow.Api/Repositories/Management/DepartmentManagement/`	`IDepartmentRepository.cs`, `DepartmentRepository.cs`
Repository (used)	`TaskFlow.Api/Repositories/Management/EmployeeManagement/Employees/`	`IEmployeeRepository.cs` (`GetByIdsAsync`, owned by 03.1)

Layer	Path	Files
DTOs	`TaskFlow.Api/Dtos/Management/DepartmentManagement/`	`DepartmentDto.cs` (`DepartmentDto`, `DepartmentWithCountsDto`, `CreateDepartmentRequest`, `UpdateDepartmentRequest`, `MemberRefDto`, `DepartmentMembersDto`, `UpdateDepartmentMembersRequest`)
Validators	`TaskFlow.Api/Dtos/Management/DepartmentManagement/Validators/`	`CreateDepartmentRequestValidator.cs`, `UpdateDepartmentRequestValidator.cs`
Entities	`TaskFlow.Api/Models/Entities/`	`Department.cs`, `DepartmentManager.cs`, `UserDepartment.cs`
Migrations	`TaskFlow.Api/Persistence/`	`20261012104823_AddDepartmentsAndUserDepartment`, `20261021070943_FixDepartmentFollowUps` (others shared, see the parent document)

4. Frontend File Structure

Type	Path	Purpose
Route / Guard	`src/app/modules/management/management.routes.ts`, `src/app/shared/auth/auth.guard.ts`	`/departments` with `permissionGuard` on `DepartmentScreenPermissions`
Menu	`src/app/layouts/nav-items.ts`	Department Management item, shown when `DepartmentScreenPermissions.some(has)`
Permission constants	`src/app/shared/auth/permissions.ts`	`DepartmentScreenPermissions` (create, edit, delete, members)
Page component	`src/app/modules/management/department-management/pages/`	`department-list.component.ts/.html`
Child components	`src/app/modules/management/department-management/components/`	`department-form-dialog.component.ts`, `department-members-dialog.component.ts/.html`, `department-members-dialog.model.ts` (`MemberSort`, `MemberOption`)
API service	`src/app/modules/management/department-management/services/department.service.ts`	CRUD and members on `/api/v1/departments`
Models	`src/app/modules/management/department-management/models/department.model.ts`	`Department`, `DepartmentWithCount`, `DepartmentMembers`, `MemberRef`
API service (used)	`src/app/modules/management/employee-management/services/user.service.ts`	People list for the members dialog (03.1)

5. API & Logic

5.1 API Endpoints

#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
1	GET	`/api/v1/departments`	All live departments with manager and employee counts	`departments.view`	–	`IReadOnlyCollection<DepartmentWithCountsDto>`
2	GET	`/api/v1/departments/{id}`	One department	`departments.view`	–	`DepartmentDto`
3	GET	`/api/v1/departments/{id}/members`	Managers and employees of the department	`departments.view`	–	`DepartmentMembersDto`
4	POST	`/api/v1/departments`	Create (201 + Location)	`departments.create`	`CreateDepartmentRequest`	`DepartmentDto`
5	PUT	`/api/v1/departments/{id}`	Rename / change description	`departments.edit`	`UpdateDepartmentRequest`	`DepartmentDto`
6	PUT	`/api/v1/departments/{id}/members`	Replace all managers and employees in one call	`departments.members.manage`	`UpdateDepartmentMembersRequest`	`DepartmentMembersDto`
7	DELETE	`/api/v1/departments/{id}`	Soft delete (only when empty)	`departments.delete`	–	`204 No Content`

5.2 Request / Response Samples

```
// POST /api/v1/departments
{ "name": "Engineering", "description": "Product engineering team" }
```

```
// 201 Created (DepartmentDto)
{ "id": 3, "name": "Engineering", "description": "Product engineering team", "createdAt":
"2026-10-31T10:00:00Z" }
```

```
// PUT /api/v1/departments/3/members
{ "managerIds": [5], "employeeIds": [14, 15] }
```

```
// 200 OK (DepartmentMembersDto)
{
  "departmentId": 3,
  "managers": [ { "id": 5, "fullName": "Ravi Menon", "username": "ravi", "email":
"ravi@taskflow.com", "isActive": true } ],
  "employees": [ { "id": 14, "fullName": "Anu Thomas", "username": "anu", "email":
"anu.thomas@taskflow.com", "isActive": true } ]
}
```

5.3 Service Layer Logic

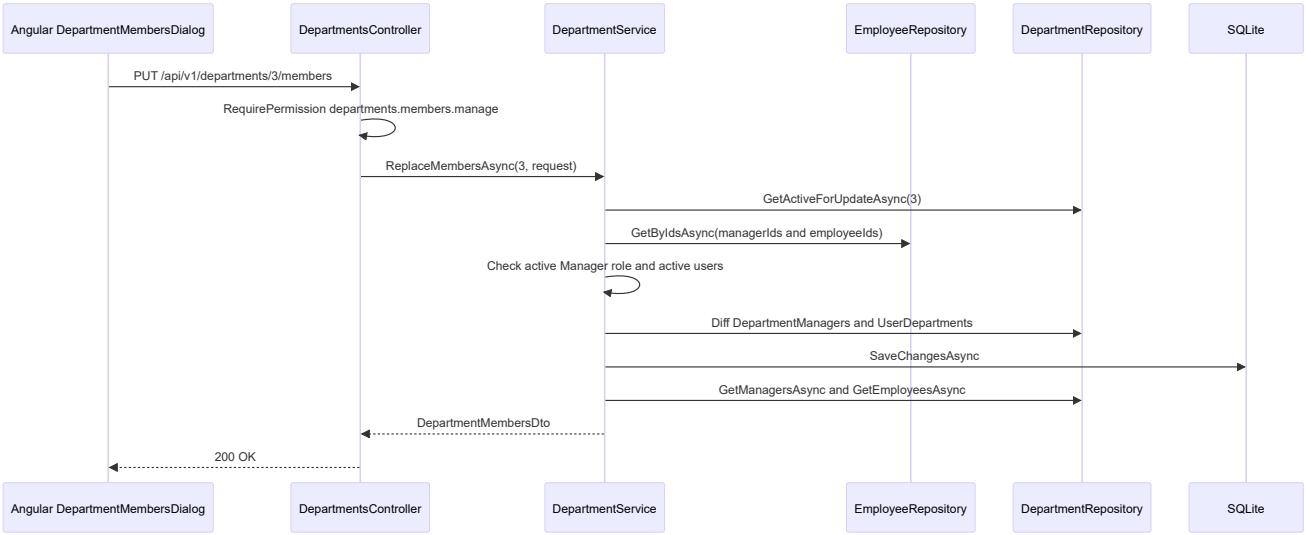
Class: `DepartmentService` (`IDepartmentRepository`, `IEmployeeRepository`, `IUnitOfWork`)

- `GetAllAsync` returns `GetAllWithCountsAsync`. `GetByIdAsync` maps with `ToDto`.
- `CreateAsync(CreateDepartmentRequest)` trims the name. If `NameTakenAsync(name, null)` is true it throws "Department name already exists.". Otherwise it adds the `Department` (`CreatedAt` = UTC now) and saves.
- `UpdateAsync(id, UpdateDepartmentRequest)` loads with `GetActiveForUpdateAsync` (null gives 404) and runs the uniqueness check excluding this id. It updates the name and description and saves.
- `DeleteAsync(id)`: `HasMembersAsync` true throws "Department cannot be deleted because it still has members. Remove them first.". Otherwise it removes the `DepartmentManagers` links, sets `IsDeleted = true` and saves.
- `GetMembersAsync(departmentId)`: returns null if the department is not live, otherwise `DepartmentMembersDto(managers, employees)`.
- `ReplaceMembersAsync(departmentId, UpdateDepartmentMembersRequest)`
    1. Returns null when the department is not live.
    2. De-duplicates the ids. Every manager id must be an existing, active user with `RoleId == SystemRoles.ManagerId`, otherwise "User {id} is not an active manager.". Every employee id must be an existing, active user, otherwise "User {id} is not an active user.".
    3. Diffs the `DepartmentManager` links and the `UserDepartment` rows for this department only.
    4. Saves and returns `GetMembersAsync`.

5.4 Repository Layer

Method	Description	Query Notes
`GetAllWithCountsAsync()`	List with counts	`AsNoTracking`. `ManagerCount` counts `DepartmentManagers` whose user is live. `EmployeeCount` counts `UserDepartments` (filtered)
`GetAsync(id)`	Read one	`AsNoTracking` (query filter hides deleted rows)
`GetActiveForUpdateAsync(id)` / `ExistsActiveAsync(id)` / `CountActiveAsync(ids)`	Live checks	`!IsDeleted`
`NameTakenAsync(name, exceptId)`	Unique name	`Name.ToLower() == name.ToLower()` among live rows
`HasMembersAsync(id)`	Delete guard	Any live manager link or any membership
`GetManagersAsync(id)` / `GetEmployeesAsync(id)`	Members as `MemberRefDto`	Ordered by FullName, including the `IsActive` flag
`GetManagerLinksForUpdateAsync` / `GetMembershipsForUpdateAsync`	Tracked rows for the diff	–
`AddManagerLink` / `RemoveManagerLinks` / `AddMembership` / `RemoveMemberships`	Stage changes	Saved by the service
`EmployeeRepository.GetByIdsAsync(ids)` (03.1)	Look up the picked people	`AsNoTracking`

5.5 Sequence Diagram



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6. Database Design

This child owns `Departments`, `DepartmentManagers` and `UserDepartments`.

Table	Column	Type	Constraints	Description
`Departments`	`Id`	int	PK, Identity	–
`Departments`	`Name`	string(200)	Not Null, Unique filtered index (`"IsDeleted" = 0`)	Department name
`Departments`	`Description`	string(500)	Nullable	–
`Departments`	`CreatedAt`	DateTime	–	–
`Departments`	`IsDeleted`	bool	Global query filter	Soft delete
`DepartmentManagers`	`ManagerUserId`, `DepartmentId`	int, int	Composite PK. FKs Cascade. Index on `DepartmentId`	Who manages the department
`UserDepartments`	`UserId`, `DepartmentId`	int, int	Composite PK. FKs Cascade. Index on `DepartmentId`. Filter hides deleted user or department	Department membership

Other tables this child reads

Table	Owner	How this child uses it
`Users`	03.1	Members are checked for `IsActive` and, for managers, `RoleId == ManagerId`. Names, usernames and emails are read into `MemberRefDto`

Relationships

- `DepartmentManagers.ManagerUserId → Users.Id` (FK, Cascade) and `DepartmentManagers.DepartmentId → Departments.Id` (FK, Cascade)

- `UserDepartments.UserId → Users.Id` (FK, Cascade) and `UserDepartments.DepartmentId → Departments.Id` (FK, Cascade)
- Because departments are soft-deleted, the cascade rarely fires. `DepartmentService.DeleteAsync` removes the manager links itself, and deletion is refused while any membership exists.

7. Security & Authorization

- Every endpoint uses `[RequirePermission(PermissionKeys.X)]`. There are no inline checks.
- **Data scope:** none. The list returns Organization-wide data to anyone holding `departments.view`.
- Authentication, the permission policy, the MFA setup gate and legacy key expansion (`departments.manage`) are described in the parent document, section 7.

Permission Key	Endpoint(s)	UI Element Controlled
`departments.view`	`GET /api/v1/departments*`	Department data (also in the employee form). Does not open the menu or route
`departments.create`	`POST /api/v1/departments`	"Add department" header button. Opens the menu and route
`departments.edit`	`PUT /api/v1/departments/{id}`	Card "Edit". Opens the menu and route
`departments.delete`	`DELETE /api/v1/departments/{id}`	Card "Delete". Opens the menu and route
`departments.members.manage`	`PUT /api/v1/departments/{id}/members`	Card "Manage members". Opens the menu and route

8. Validations & Error Handling

Rule	Layer	HTTP Code	Error Message
Department `Name` empty or over 200, `Description` over 500	`CreateDepartmentRequestValidator`, `UpdateDepartmentRequestValidator` via `ValidationFilter`	400	FluentValidation default messages
Duplicate department name	`DepartmentService`	400	Department name already exists.
Department has members	`DepartmentService.DeleteAsync`	400	Department cannot be deleted because it still has members. Remove them first.
Bad department member	`DepartmentService.ReplaceMembersAsync`	400	User {id} is not an active manager. / User {id} is not an active user.
Department not live	Controller (null from service)	404	Empty body

Shared errors (409 unique race and concurrency, 403 policy and MFA gate, 500) are in the parent document, section 8.

9. Integrations / Side Effects

- **Notifications / Email:** none.
- **Activity Log:** none (see the parent document).

- **Manager oversight:** department deletion removes the `DepartmentManagers` links. Department managers feed `IEmployeeRepository.GetTeamMemberIdsAsync` (manager's team for task visibility and the manager dashboard) and the `managers` / `reports` shown in `UserDto`.
- **AI Assistant:** no tools. The help topic `department-management.md` describes this screen (see section 12).

10. Configuration

Not applicable – Department Management has no settings of its own. The shared `ConnectionStrings:Default` is in the parent document.

11. Testing

Test Type	File / Location	Covers
Integration	`tests/TaskFlow.IntegrationTests/Management/DepartmentManagement/DepartmentsEndpointsTests.cs`	`/api/v1/departments` create, rename, delete and `PUT /{id}/members`: Administrator only (Manager can only view, Employee 403, anonymous 401), duplicate names refused ignoring case, blank and over-200-character names 400, 404 for a missing department, members replaced wholesale (an invalid manager or inactive employee rejects the whole replace), delete blocked while members exist, a deleted manager's link no longer blocks delete, a deleted department's name can be reused
Unit	`tests/TaskFlow.UnitTests/Management/DepartmentManagement/DepartmentRequestValidatorsTests.cs`	Create and update validators: name required and at most 200 characters, description at most 500 characters
Unit	`tests/TaskFlow.UnitTests/Persistence/DepartmentFollowUpMigrationTests.cs`	Migrations after the Department/Project split: custom roles holding `departments.*` also get the `projects.*` keys, and a deleted department or project no longer reserves its name (shared with 03.3)
Frontend unit	`src/app/modules/management/department-management/pages/department-list.component.spec.ts`	Header count and cards with member counts, client-side name search and no-match message, add and edit dialogs, members dialog that reloads the list, delete guarded while members or managers remain, delete with confirmation and toast, empty and error states
Frontend unit	`src/app/modules/management/department-management/components/department-form-dialog.component.spec.ts`	Add and edit dialog: "Name is required", server error kept in the dialog for a duplicate name, update closes, toasts and emits saved

Test Type	File / Location	Covers
Frontend unit	`src/app/modules/management/department-management/components/department-members-dialog.component.spec.ts`	Members dialog: loads members and pickers in parallel, managers and employees in one list with role tags, inactive members kept, search, sort, select and clear all, sends managers and employees separately, load and save errors
Frontend unit	`src/app/modules/management/department-management/services/department.service.spec.ts`	`DepartmentService` calls to `/api/v1/departments` (list, get, create, update, delete, get and replace members) and error propagation

12. Known Limitations / Gaps found

- **BRD 06 item 3:** "Display employees associated with each department" is met only through counts and the Manage members dialog. There is no read-only member view for users without `departments.members.manage`.
- **Frontend gating gaps (confirmed in code):**
  - The department empty-state **Add department** button is shown without checking `departments.create`.
  - `departments.view` has no frontend reference.
- **Hard-coded roles:** the department members dialog lists only the `Manager` and `Employee` roles, so custom-role users and Administrators can never be picked, although the API accepts any active user as an employee. Only the Manager system role can be a department manager (see the parent document).
- **Extra permission needed silently:** the department members dialog needs `employees.view` to list people.
- Department search happens on the client and the list is not paged (see the parent document).
- **Outdated help knowledge:** `Services/Assistant/Knowledge/department-management.md` is titled "Project management" and links to `/departments`.

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